sci

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

In Space Age, find out all about crowdsourced astronomy, learn about the Ramanujan Machine in R&D, then find out how dogs were domesticated in Ignorance, and more....

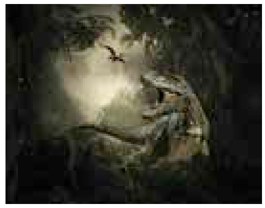
ENIAC turns 75

The world's first general purpose computer turned 75 earlier this month, find out its fascinating history and the untold story of the "Eniac 6" who programmed the machine.
http://dgit.in/eniac75

WHAT'S NEW

Dust settles on dino extinction

There has been increasing scrutiny on the narrative that a single asteroid impact wiped out the dinosaurs around 66 million years ago in recent times. There is plenty of evidence to suggest that excessive volcanism in the Deccan traps contributed to, or exacerbated the adverse environmental effects caused by the Chicxulub event. One of the findings that provided additional evidence in support of the meteor impact theory was a global spike in iridium in the geological area around the time of the impact. Iridium is an element that is uncommon on Earth, but relatively common in meteorites. One of the criticisms of this finding was that the source of the iridium may not have been the

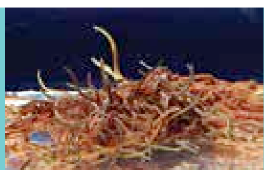


meteorite. This criticism has now been settled by researchers who investigated the impact crater, and found iridium in the crater. The team was also able to date the iridium deposits down to a period of two decades after the impact. In addition to the iridium, the study also found high concentrations of sulphur. The finding paints a picture of pulverised rock, meteor dust and sulfur compounds blown into the atmosphere, that blocked out the Sun for two decades.
http://dgit.in/dinodust

Exploring teamwork among lions

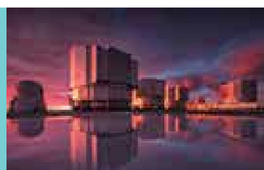
Researchers conducted a long term study on the behaviors of male lions in Gir, to better understand how male lions group up, and form teams. Typically, a single dominant alpha male monopolises all the breeding opportunities in a pride. Male members in a pride have to move out to breed, and displace the dominant male in another pride. It is not easy for a single male lion to displace a

dominant male, so male lions often form a coalition with one or more male partners to increase their chances. In most cases, the teams of unrelated males consisted of two lions, though related males banded together in larger teams. The researchers calculated fitness scores for the lions and dominance rankings between the individuals. The team also selected samples of 23 adult lions in 10 coalitions, ranging in size from two to four individuals. The research showed that single lions had the lowest fitness scores, while pairs fared better, and larger coalitions had larger collective fitness scores.
http://dgit.in/girlion



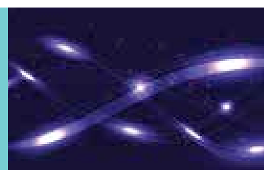
Robot blobs

Unremarkable worms that form fish food, can clump up together to form balls that offer each individual collective protection. Researchers are looking at how to get robot swarms to behave in similar ways, so that the lessons learned from nature can be applied in future robotics.
http://dgit.in/blobot



Imaging exoplanets

Researchers have developed a new technique that combines infrared imaging with extremely long exposures, that has the potential to directly capture images of nearby exoplanets. The research suggests a ten fold increase in current capabilities of directly imaging exoplanets.
http://dgit.in/exop10



Quantum magnonics

One of the challenges of quantum computing has been identifying suitable subatomic particles, and researchers have now managed to control the interactions of microwave photons and magnons for the first time, which could be used to produce novel quantum computing devices.
http://dgit.in/quamag